



Nonequilibrium thermodynamics of the Markovian Mpemba effect and its inverse

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Under certain conditions, it takes a shorter time to cool a hot system than to cool the same system initiated at a lower temperature. This phenomenon—the “Mpemba effect”—was first observed in water and has recently been reported in other systems. Whereas several detail-dependent explanations were suggested for some of these observations, no common underlying mechanism is known. Using the theoretical framework of nonequilibrium thermodynamics, we present a widely applicable mechanism for a similar effect, the Markovian Mpemba effect, derive a sufficient condition for its appearance, and demonstrate it explicitly in three paradigmatic systems: the Ising model, diffusion dynamics, and a three-state system. In addition, we predict an inverse Markovian Mpemba effect in heating: Under proper conditions, a cold system can heat up faster than the same system initiated at a higher temperature. We numerically demonstrate that this inverse effect is expected in a 1D antiferromagnet nearest-neighbors interacting Ising chain in the presence of an external magnetic field. Our results shed light on the mechanism behind anomalous heating and cooling and suggest that it should be possible to observe these in a variety of systems.

nonequilibrium thermodynamics | Mpemba effect | Ising model | Markov dynamics

Consider cooling a system initiated at a hot temperature by coupling it to a cold bath. Intuitively, we expect that the cooling time should increase with the system’s initial temperature. Surprisingly, this is not always the case: More than 2,300 years ago Aristotle documented that the inhabitants of Pontus were using hot water, instead of cold water, to prepare ice rapidly (1). Nowadays, anomalous cooling in water or other substances is referred to as the Mpemba effect: When two samples of the same substance identical in all macroscopic parameters except for their initial temperatures are simultaneously cooled by the same cold bath, it can take less time to cool the sample initiated at a higher temperature.

The exact mechanism behind anomalous cooling in water and even its existence (2, 3) are still under debate. Clearly, this effect contradicts Newton’s heat law, where the rate of change of the system’s temperature is simply proportional to the temperature difference between the system and the bath. When the cooling process is quasistatic, namely when the system’s temperature follows Newton’s heat law, the hot system necessarily lags behind the cold system; i.e., the Mpemba effect cannot occur. On the other hand, this effect has been observed in water (4) and more recently in several other substances, e.g., nanotube resonators (5), magneto-resistance alloys (6), clathrate hydrates (7), and granular systems (8).

The cooling process in the Mpemba effect, i.e., quenching, is in general not quasistatic, but rather a genuine far-from-equilibrium process. Several aspects of nonequilibrium cooling have been considered to partially explain the Mpemba effect in water: hydrodynamic effects (9), supercooling (10, 11), evaporation (12, 13), impurities in the water sample (14, 15), and even the microscopic structure of the hydrogen bond (16, 17). Whereas these mechanisms explain some of the observations in

water and clathrate hydrates, they are all substance specific and thus cannot explain the Mpemba effect observed in other substances, e.g., in magneto-resistance alloys or granular systems.

In this paper, we consider anomalous cooling processes in the general framework of nonequilibrium statistical mechanics. For systems undergoing Markovian dynamics, we provide a sufficient condition, accompanied with heuristic intuition for its appearance. We stress in advance that it is not clear that the proposed theory is the dominant mechanism in the specific phenomenon observed in water, and to distinguish between the two we refer to the former as the Markovian Mpemba effect. To illustrate the Markovian Mpemba effect, we first demonstrate it in a minimal three-state system and then in one of the most important models in statistical mechanics—the Ising model. We show that the Markovian Mpemba effect can be found in an antiferromagnetic nearest-neighbor interacting Ising spin chain, in the presence of an external magnetic field. We then predict the existence of an inverse Markovian Mpemba effect that occurs during heating: When two systems are heated by the same hot bath, it can take a shorter time to heat the initially colder system. This inverse effect is demonstrated on both the three-state system and the antiferromagnetic Ising chain.

This paper is structured in the following order. In *Theory*, we set up the framework to describe cooling processes in Markovian dynamics, introduce the distance-from-equilibrium function, discuss its properties, and use it to precisely define the Markovian Mpemba effect. These ideas are then illustrated by a minimal model of the Markovian Mpemba effect—a three-state Markov system. In *Results* we analytically obtain a sufficient condition for the Markovian Mpemba effect and demonstrate it for the three-state system as well as for the antiferromagnetic 1D Ising chain. An additional result is the prediction of an inverse

Significance

It is commonly expected that cooling a hot system takes a longer time than cooling an identical system initiated at a lower temperature. Surprisingly, this is not always the case; in various systems, including water and magnetic alloys, it has been observed that a hot system can be cooled faster. These anomalous cooling effects are referred to as “the Mpemba effect”, and so far they lack a generic details-independent explanation. Based on recent developments in the theory of nonequilibrium thermodynamics, we propose a generic mechanism for similar effects, demonstrate it in various systems, and predict a similar anomalous behavior in heating.

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Markovian Mpemba effect, which is demonstrated using similar systems. In *Discussion: Energy Landscape and the Markovian Mpemba Effect*, we provide some physical intuition for the Markovian Mpemba effect based on certain geometric properties of the system's energy landscape. This intuition is numerically demonstrated by an example of diffusive dynamics. And finally, we present *Conclusions*.

Theory

Markovian Dynamics. In the absence of a thermal bath, any isolated classical system—be it a glass of water or a magnetic alloy—evolves deterministically in its phase space in accordance with its Hamiltonian. (We neglect in this discussion any quantum source of nondeterminism.) However, when coupled to a thermal bath, the dynamic of the system is no longer deterministic due to the random thermal fluctuations. It is instructive to describe systems coupled to a thermal bath by a probability distribution $\vec{p}$, which quantifies the probability to find the system in any microstate—a point in its configuration space (or phase space). With an ideal thermal bath (namely a bath with zero memory), the dynamic of $\vec{p}$ is Markovian and follows the equation

$$\frac{d\vec{p}(t)}{dt} = \mathcal{L}_{T_b} \vec{p}(t), \quad [1]$$

where $\mathcal{L}_{T_b}$ is a linear Markovian operator that is determined by the system itself, the bath temperature T_b , and the coupling between the system and the bath. $\mathcal{L}_{T_b}$ is assumed to be ergodic and to satisfy detailed balance, so that the system has a unique steady state, which is the thermal equilibrium (the Boltzmann distribution corresponding to the bath temperature T_b). Dynamics with the form of Eq. 1 are widely used to model a variety of thermal processes, e.g., the Kramer–Fokker–Planck dynamic for a Brownian particle, the Glauber dynamics of a classical spin system, and the Lindblad dynamics of a quantum system.

To simplify the presentation, we first focus on systems with a finite number of states, although a similar analysis can be carried out for systems with a continuous state space (example in *Discussion: Energy Landscape and the Markovian Mpemba Effect*). Let us denote the probability to find a system at the i th state at time t by $p_i(t)$ and denote the Markovian operator $\mathcal{L}_{T_b}$ by the transition rate matrix $R(T_b)$. The rate of transition from state j to state i , $R_{ij}(T_b)$, is jointly determined by the energy (E_j), the energy barrier between the states (B_{ij}), and the temperature of the thermal bath (T_b). The system thus evolves according to

$$\frac{dp_i(t)}{dt} = \sum_j R_{ij}(T_b) p_j(t) \quad \text{for } i = 1, 2, \dots, n. \quad [2]$$

The elements $R_{ij}(T_b)$ take the Arrhenius form (18)

$$R_{ij}(T_b) = \begin{cases} \Gamma e^{-\frac{B_{ij}-E_j}{k_B T_b}} & i \neq j \\ -\sum_{k \neq i} R_{ki} & i = j \end{cases}, \quad [3]$$

where Γ is a rate constant that fixes the dimensions, $B_{ij} = B_{ji}$, and the diagonal terms R_{ii} are defined such that the normalization, $\sum_i p_i(t) = 1$, is conserved. Under this dynamic, any initial distribution eventually relaxes into the equilibrium (Boltzmann) distribution

$$\pi_i(T_b) = \frac{e^{-E_i/k_B T_b}}{\sum_i e^{-E_i/k_B T_b}}. \quad [4]$$

The Distance Function. In most previous descriptions of the Mpemba effect, the cooling rate was characterized by measuring the time it takes for the water to freeze (this is a very subtle definition, discussion in ref. 4) or by tracking the readout of

a thermometer (19). However, the former cannot be applied to systems without a phase transition, although anomalous cooling was observed in such systems, e.g., in granular materials far from the jamming transition. The readout of a thermometer, on the other hand, might depend on specific details such as the sensor's exact location and its working principle.

To generalize the discussion on anomalous cooling and avoid these issues, we quantify the rate of cooling by constructing a distance-from-equilibrium function, $D[\vec{p}(t); T_b]$, and observing its decay over time. $D[\vec{p}; T_b]$ measures the distance of any distribution $\vec{p}$ from the ultimate equilibrium Boltzmann distribution $\vec{\pi}(T_b)$ given in Eq. 4. Given a distance-from-equilibrium function we can define the Markovian Mpemba effect as follows: If there exist three temperatures, $T_b < T_c < T_h$ such that two systems that are initially prepared at a hot and a cold equilibria [$\vec{p}^h(0) = \pi(T_h)$ and $\vec{p}^c(0) = \pi(T_c)$] relax toward the same equilibrium $\pi(T_b)$ according to Eq. 2, and if there exists some finite time t_m such that $D[\vec{p}^h(t); T_b] < D[\vec{p}^c(t); T_b]$ for all $t > t_m$, namely the distance from equilibrium of the initially hot system becomes smaller than that of the cold system, then the Markovian Mpemba effect exists in the system.

There are many reasonable choices for a distance-from-equilibrium function. However, the Markovian Mpemba effect may be falsely reported due to a poor choice of the distance function $D[\vec{p}; T_b]$. To avoid such cases, we demand that $D[\vec{p}; T_b]$ satisfies the following three properties: (i) When the system relaxes toward its thermal equilibrium (from any initial distribution), the value of $D[\vec{p}(t); T_b]$ is always monotonically nonincreasing with time—the relaxation process cannot push the system away from equilibrium. (ii) The distance from equilibrium of a system initiated at temperature T_h is greater than that of the same system initiated at T_c when $T_b < T_c < T_h$. In other words, $D[\vec{\pi}(T); T_b]$ is a monotonically increasing function of T for any $T > T_b$. And finally, (iii) the distance from equilibrium is a continuous, convex function of $\vec{p}$. As we show below, the identification of the Markovian Mpemba effect is indifferent to the specific choice of the distance function, as long as these three conditions are all satisfied. [Alternative commonly used distance functions that satisfy all of the three requirements are the Kullback–Leibler divergence and the L_1 distance between $\vec{p}$ and $\vec{\pi}(T_b)$.] In what follows we use the entropic distance between any distribution $\vec{p}(t)$ and the equilibrium distribution $\vec{\pi}(T_b)$, defined by the total amount of entropy production (of the system and the bath) in the relaxation starting from $\vec{p}$ and ending at $\vec{\pi}(T_b)$ (see *SI Appendix, section I* for a derivation),

$$D_e[\vec{p}; T_b] = \sum_i \left(\frac{E_i \Delta p_i}{T_b} + p_i \ln p_i - \pi_i^b \ln \pi_i^b \right), \quad [5]$$

where π_i^b denotes the equilibrium probability at temperature T_b , $\Delta p_i = p_i - \pi_i^b$, and the units are set such that $k_B = 1$. $D_e[\vec{p}; T_b]$ is a continuous convex function of $\vec{p}$, and by the second law of thermodynamics $D_e[\vec{p}(t); T_b]$ is a monotonically decreasing function of time (see ref. 20 for a proof). Moreover, as we show in *SI Appendix, section II*, $D_e[\vec{\pi}(T); T_b]$ is a monotonic function of T for $T > T_b$. Therefore, this choice of $D[\vec{p}; T_b]$ satisfies all of the three requirements for the distance function.

Example: A Three-State Model. To illuminate the discussion, we next introduce a minimal model that shows the Markovian Mpemba effect—a three-state system (Fig. 1B) that is coarse grained from an overdamped system with a three-well energy landscape (Fig. 1A). In this specific example, the energies of the three states are given by $E_1 = 0$, $E_2 = 0.1$, and $E_3 = 0.7$ and the energy barriers by $B_{12} = 1.5$, $B_{13} = 0.8$, and $B_{23} = 1.2$. The space of probabilities for this three-state system contains all of the triplets (p_1, p_2, p_3) that satisfy $0 \leq p_i \leq 1$ and the normalization condition $p_1 + p_2 + p_3 = 1$. As shown in Fig. 1C, this space forms

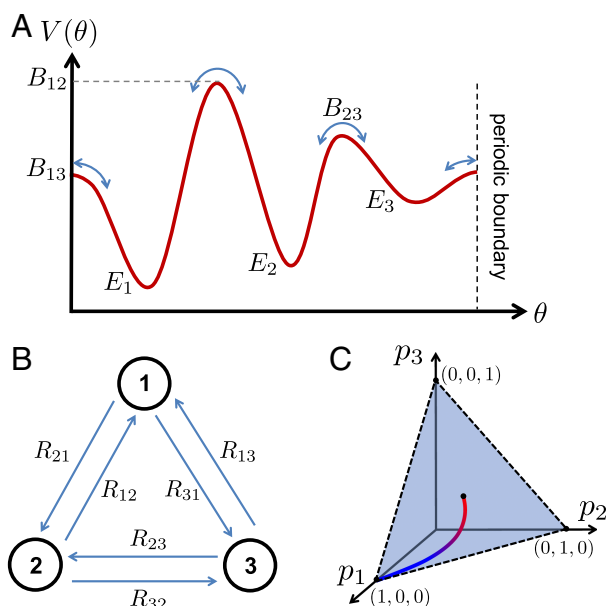


Fig. 1. (A) Sketch of an energy landscape with three metastable states. The energy of each state is denoted by E_i , and the barrier between two states is denoted by B_{ij} . (B) An effective description of A as a three-state system, where R_{ij} denotes transition rate from j to i and is given by Eq. 3. (C) The probability distribution among the three states can be described by the vector $\vec{p} = (p_1, p_2, p_3)$, and all possible values of $\vec{p}$ form a simplex in (p_1, p_2, p_3) (shaded triangle). The curved line is the quasistatic locus, namely the set of Boltzmann distributions $\vec{\pi}(T)$ corresponding to different temperatures from 0 (blue end) to ∞ (red end).

a 2D simplex whose vertices are located at $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$. Any initial probability distribution—a point in the simplex—evolves according to the master equation toward the equilibrium point, $\vec{\pi}(T_b)$.

For each temperature T , the Boltzmann distribution $\pi(T)$ given by Eq. 4 is a point in the triangle. The set of Boltzmann distributions for all temperatures $T \in (0, \infty)$ forms a curve, known as the quasistatic locus (21) (Fig. 1C). In quasistatic cooling, both the hot and the cold systems evolve along the quasistatic locus toward the equilibrium, and hence the hot system lags behind the cold system. However, this is not the case in a nonequilibrium cooling process. By solving the master equation, we show in Fig. 2 that the hot system and the cold system evolve according to two nonoverlapping paths $[\vec{p}^h(t)$ and $\vec{p}^c(t)]$ toward the same final point $\vec{\pi}(T_b)$. The Markovian Mpemba effect occurs in this case because the distance along the trajectory of the hot system decays faster than along the trajectory of the cold system, as shown in Fig. 2, *Inset*.

Results

Sufficient Condition for the Markovian Mpemba Effect. The Markovian Mpemba effect occurs when the distance from equilibrium of the initially hot system is larger than that of the initially cold system at $t=0$, but becomes smaller after a long enough time. To find the conditions under which the Markovian Mpemba effect occurs, we note that the asymptotic behavior of $\vec{p}(t)$ at long times ($t \rightarrow \infty$) is dictated by the two largest eigenvalue eigenvectors of R . As we will show, it is the coefficient of the second largest eigenvalue eigenvector that is responsible for the Markovian Mpemba effect. More precisely, the trajectories $\vec{p}^h(t)$ and $\vec{p}^c(t)$ can be described using the ordered eigenvalues $\lambda_1 > \lambda_2 \geq \lambda_3 \geq \dots \geq \lambda_n$ and corresponding right eigenvectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ of the rate matrix R_{ij} . Because R_{ij} satisfies detailed balance, it is diagonalizable, all its eigenvalues are real

(22, 23), and $0 = \lambda_1 > \lambda_2$, where λ_2 is the slowest relaxation rate of the system. The null eigenvector $\vec{v}_1$ is equal to the unique equilibrium distribution $\vec{\pi}(T_b)$. We can express any initial probability distribution as a linear combination of the eigenvectors, $\vec{p}(0) = \vec{\pi}(T_b) + a_2 \vec{v}_2 + \dots + a_n \vec{v}_n$. The probability evolves in time according to

$$\vec{p}(t) = \vec{\pi}(T_b) + e^{\lambda_2 t} a_2 \vec{v}_2 + \dots + e^{\lambda_n t} a_n \vec{v}_n; \quad [6]$$

namely the coefficients of the eigenvectors (except the equilibrium) decay exponentially. When λ_2 is strictly larger than λ_3 ($\lambda_2 > \lambda_3$) and $|a_2^c| > |a_2^h|$, the Markovian Mpemba effect occurs. This can be seen from the following argument: For large enough t , the terms $e^{\lambda_k t} a_k \vec{v}_k$ for $k \geq 3$ are exponentially smaller than $e^{\lambda_2 t} a_2 \vec{v}_2$. Therefore, they can be neglected, and the two relaxation trajectories can be written as $\vec{p}^h(t) \approx \vec{\pi}(T_b) + a_2^h \vec{v}_2 e^{\lambda_2 t}$ and $\vec{p}^c(t) \approx \vec{\pi}(T_b) + a_2^c \vec{v}_2 e^{\lambda_2 t}$. Hence, at large time the relaxation trajectories $\vec{p}^c(t)$ and $\vec{p}^h(t)$ follow approximately the same trajectory, and because $|a_2^c| > |a_2^h|$, $\vec{p}^c(t)$ lags behind $\vec{p}^h(t)$. By the convexity and monotonicity of $D[\vec{p}(t); T_b]$ this implies that the Markovian Mpemba effect occurs (see *SI Appendix, section III* for a proof).

Using the condition $|a_2^c| > |a_2^h|$, we next identify whether a system has the Markovian Mpemba effect at a given T_b . This can be done by considering the initial conditions along the quasistatic locus $\vec{\pi}(T)$. Each point along the quasistatic locus can be written as $\vec{\pi}(T_b + \Delta T) = \vec{\pi}(T_b) + \sum_i a_i(\Delta T) \vec{v}_i$. If at some ΔT the magnitude of $a_2(\Delta T)$ decreases as a function of ΔT , then by the above argument the Markovian Mpemba effect occurs. This condition has a simple geometric interpretation for the three-state system: The magnitude of the coefficient $a_2(\Delta T)$ initially increases for small enough ΔT . If it also decreases, then there must be a point T_m for which $\frac{da_2}{dT} \Big|_{T_m} = 0$. This means that the

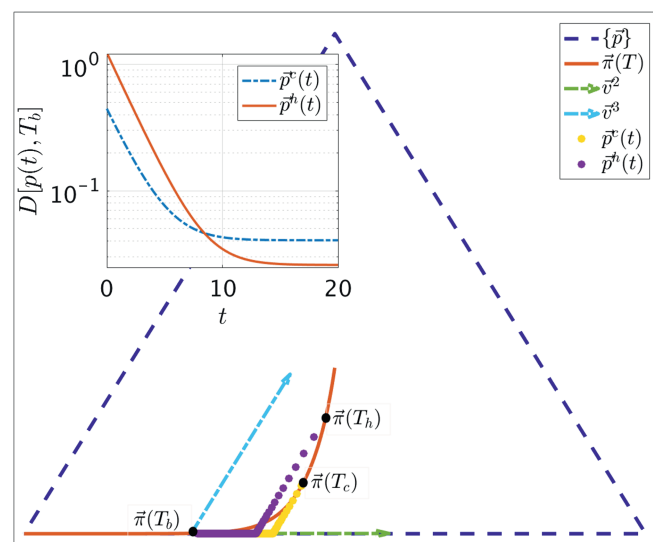


Fig. 2. The set of all normalized probability distributions forms a simplex—the dashed blue triangle. The red solid curve is the quasistatic locus, $\vec{\pi}(T)$. The dashed arrows are the two right eigenvectors of R_{ij} , $\vec{v}_2$ and $\vec{v}_3$, associated with fast (blue) and slow (green) relaxation modes. The dotted lines represent the relaxation process, $\vec{p}^c(t)$ starting at $T_c = 0.42$ (orange) and $\vec{p}^h(t)$ starting at $T_h = 1.3$ (purple). Note that the tangent to the quasistatic locus is parallel to $\vec{v}_3$ at about $\vec{\pi}(T_c)$, and therefore the coefficient of $\vec{v}_2$ decreases with the temperature and the Markovian Mpemba effect occurs. *Inset* shows the distances $D_e[\vec{p}^h(t); T_b]$ and $D_e[\vec{p}^c(t); T_b]$ both decrease with time. The initially hot system starts at a larger distance, but after some time its distance from equilibrium is smaller than that of the initially cold system.

tangent to the quasistatic locus at T_m has no contravariant coefficient along $\vec{v}_2$, or in other words, the tangent to the quasistatic locus at this point is parallel to $\vec{v}_3$ (example in Fig. 2). (The eigenvectors of R are, in general, nonorthogonal, and hence we need to differentiate between contravariant and covariant coefficients.)

Let us demonstrate this condition for the three-state system example. A sufficient condition for the Markovian Mpemba effect is that the tangent to the quasistatic locus is parallel to $\vec{v}_3$ at some temperature $T_m > T_b$. For $T_b = 0.1$, the fast relaxation vector $\vec{v}_3$ (the blue arrow in Fig. 2) is parallel to the quasistatic locus tangent at $T_m \approx 0.418$; namely the magnitude of the coefficient in the slow direction, $a_2(T)$, decreases beyond this point. Hence, for any two choices of the initial temperatures T_h, T_c above T_m , the Markovian Mpemba effect can be observed. This is verified for $T_c = 0.42$ and $T_h = 1.3$ (Fig. 2). For this system, $D_e[\vec{p}(t), T_b]$ can be analytically obtained and is plotted in Fig. 2, *Inset*. After $t_m \approx 9$, the distance from equilibrium of the initially hot system drops below that of the initially cold system, $D_e[\vec{p}^h(t), T_b] < D_e[\vec{p}^c(t), T_b]$ and hence the Markovian Mpemba effect occurs.

The Markovian Mpemba Effect in the Ising Model. To show that the discussed effect is not the result of a careful tuning of parameters, we next demonstrate the Markovian Mpemba effect in the Ising model, in which the number of parameters is significantly smaller than the number of microstates (*SI Appendix, section V*). We consider a 1D chain of N spins. Each spin can take a value $s_k \in \{1, -1\}$. The number of microstates in the system is thus 2^N . The interactions between the spins are nearest-neighbors antiferromagnetic with strength J , and there is a homogeneous external magnetic field denoted by h . The boundary conditions are given by $s_0 = s_{N+1} = 1$ (Fig. 3). The energy of any microstate is given by

$$E_i = -J \sum_{k=0}^N s_k s_{k+1} - h \sum_{k=1}^N s_k. \quad [7]$$

The transition rate between state j and state i is given by ($i \neq j$):

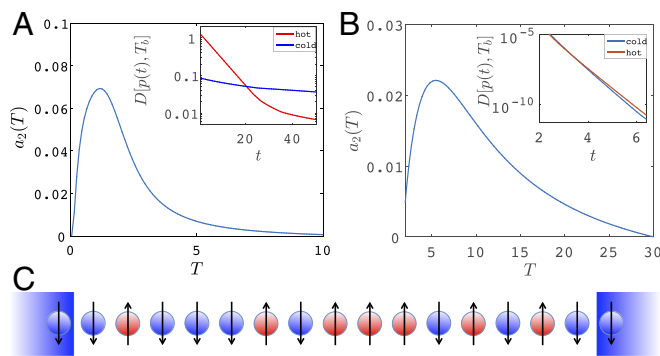


Fig. 3. The Markovian Mpemba effect and its inverse in the Ising model. As illustrated, we consider a 1D chain of 15 spins with antiferromagnetic nearest-neighbors interactions and fixed boundary conditions, in the presence of an external magnetic field. (A) The coefficient $a_2(T)$ of the slowest dynamics along the quasistatic locus as a function of the temperature, for the Markovian Mpemba effect in the Ising model. (A, *Inset*) The distance from equilibrium as function of time, for the hot (red) and cold (blue) initial conditions. (B) The coefficient $a_2(T)$ of the slowest dynamics along the quasistatic locus as a function of the temperature, for the inverse effect in the Ising model. (B, *Inset*) The distance from equilibrium as a function of time, for the hot (red) and cold (blue) initial conditions. (C) An illustration of a 1D Ising chain of 15 spins with fixed boundary conditions.

$$R_{ij}(T_b) = \begin{cases} \Gamma e^{-\frac{E_i - E_j}{2k_B T_b}} & \text{if } i \text{ and } j \text{ are different by a single spin flip.} \\ 0 & \text{if } i \text{ and } j \text{ are different by more than one spin flip.} \end{cases} \quad [8]$$

In other words, only a single spin flip is allowed at each instance of time, and the rate of any spin flip is proportional to the exponent of the energy differences between the two corresponding microstates.

Specifically, we used $N = 15$, and thus the number of microstates (this is the largest number of spins for which we could calculate $D_e[\vec{p}(t), T_b]$ and $a_2(T)$ exactly) is $2^{15} = 32,768$. The antiferromagnet coupling constant is given by $J = -2$ and the external field by $h = 0.2$. With this combination of parameters, the system has several metastable states characterized by slow transitions between them. In addition, the majority of the state space corresponds to high-energy microstates that can rapidly relax toward the minimal-energy microstate, in the presence of the magnetic field (*Discussion: Energy Landscape and the Markovian Mpemba Effect*). The temperature of the bath is $T_b = 0.05$, and the hot and cold initial temperatures are $T_h = 4.25$ and $T_c = 1.15$. Fig. 3A shows the coefficient of the slowest decay rate of systems starting from a point on the quasistatic locus as a function of the temperature, $a_2(T)$. It is nonmonotonic as we expect from a system with the Markovian Mpemba effect: Starting at $a_2(T_b) = 0$, it increases with T , reaches the maximum at $T_m \approx 1$, and starts to decrease beyond T_m . Our choice of initial temperatures satisfies the conditions for the Markovian Mpemba effect $|a_2^h| < |a_2^c|$, and we should observe the Markovian Mpemba effect in this example. Indeed, as the numerical calculation in Fig. 3A, *Inset* shows, the distance from equilibrium of the initially hot system drops below that of the initially cold system after time $t_m \approx 20$ and the effect occurs.

The Inverse Markovian Mpemba Effect. So far we have considered anomalous cooling processes. Next, we predict the existence of a similar effect for heating processes, where an initially cold system (T_c) heats faster than an initially hot system (T_h), when both are heated by the same hot bath ($T_b > T_h > T_c$). We first illustrate this effect by a similar three-state system, but with different parameters: $E_1 = 0$, $E_2 = 0.1$, $E_3 = 1$, $B_{12} = 2$, $B_{13} = 1.01$, $B_{23} = 10$. The cold system starts at the equilibrium distribution corresponding to the temperature $T_c = 0.3$ and the hot system at the equilibrium distribution corresponding to $T_h = 0.78$. For both systems, the bath temperature is $T_b = 10$. The corresponding dynamics are shown in Fig. 4, and they clearly show the inverse Markovian Mpemba effect. A sufficient condition for the inverse Markovian Mpemba effect is $\lambda_2 > \lambda_3$ and $|a_2^h| > |a_2^c|$. This can be derived from a similar argument to that used for the direct effect: $|a_2^h| > |a_2^c|$ implies that for large enough t , $\vec{p}^h(t)$ follows almost the same trajectory but lags behind $\vec{p}^c(t)$, and thus it takes a longer time to heat the initially hot system.

Interestingly, this inverse effect can be found in a similar 1D antiferromagnetic Ising chain, for different parameters. The $N = 15$ spin system with $J = -4$, $h = 8.5$, and $T_b = 30$ is numerically shown to satisfy the condition for the inverse Markovian Mpemba effect (nonmonotonic curve in Fig. 3B). This prediction is verified by choosing the two initial temperatures to be $T_h = 2.4$ and $T_c = 1.8$ and calculating their distance function $D_e[\vec{p}^{h,c}(t), T_b]$ as a function of time (Fig. 3B, *Inset*).

Discussion: Energy Landscape and the Markovian Mpemba Effect

To gain some physical intuition into the effect, we next discuss its relationship with certain geometric properties of the system's energy landscape. A common feature for systems that demonstrate the Markovian Mpemba effect is that the energy

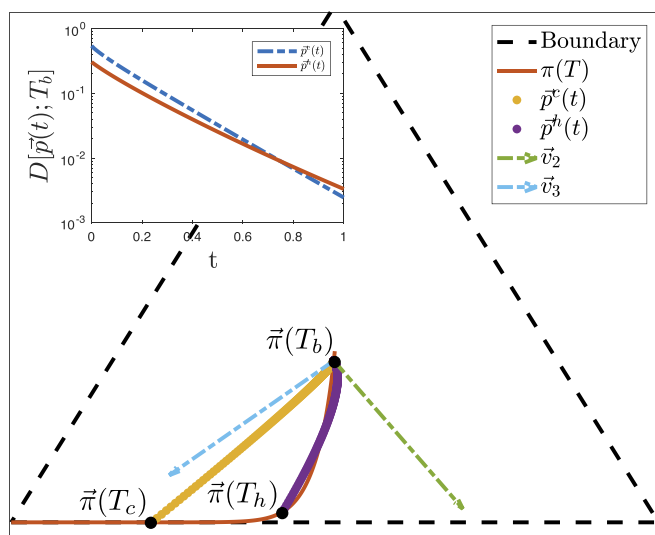


Fig. 4. The inverse Markovian Mpemba effect. Two identical systems, initiated at a cold temperature T_c and a hot temperature T_h are coupled to a hot bath with temperature $T_b > T_h > T_c$. The initially cold system has a very small component in the slow relaxation mode $\vec{v}_2$ (green dashed arrow), whereas the initially hot system has a much larger coefficient in the slow direction (even though it has a smaller distance from the equilibrium), and hence it decays slower than the initially cold system. (Inset) The distance from equilibrium for both systems as a function of time. The initially cold system decays faster and the inverse Markovian Mpemba effect occurs.

landscape defined on their microstates has multiple local minima or metastable energy wells (therefore we used the antiferromagnetic, rather than the ferromagnetic 1D Ising model). These are schematically illustrated in Fig. 5A. The relaxation rate from a metastable state into the global minimum is slow, whereas the relaxation rates from the high-energy states into the global minimum are fast (arrows in Fig. 5A). The equilibrium distribution corresponding to T_b is mostly concentrated in the lowest-energy state. At a cool initial temperature T_c , a significant portion of the corresponding equilibrium probability is “trapped” in the metastable states, and hence the relaxation from the equilibrium of T_c to that of T_b is slow. In contrast, at the high temperature T_h the energy landscape plays only a minor role in the equilibrium distribution—most of the probability is spread across the high-energy microstates, which flow into the global minimum through a fast relaxation.

To illustrate this argument, we construct an example of Fokker–Planck dynamics in the 1D potential landscape shown in Fig. 5B. At high temperature T_h , the Boltzmann distribution spreads almost uniformly over the configuration space. At low temperature T_c , the Boltzmann distribution is concentrated at the two energy wells, and at the bath temperature T_b most of the probability is confined in the deepest well (Fig. 5C). When coupled to the cold bath, the high-energy configurations in the initially hot system rapidly flow into the lowest-energy well, whereas in the initially cold system the probability in the left metastable energy well takes a long time to relax into the lowest-energy well. In other words, the relaxation of the cold initial condition is slow because it requires some probability to transfer across a barrier from the metastable well into the lowest well, whereas in the initially high-temperature system only a small amount of probability has to cross the barrier when relaxing toward the final equilibrium, and the relaxation is thus faster. This argument is numerically verified by calculating the distance from equilibrium function of the hot and the cold systems—these are plotted in Fig. 5C, Inset. See SI Appendix, section IV for a detailed discussion of the example.

Conclusions

In this paper we have discussed a Markovian Mpemba effect in cooling and predicted a similar effect in heating (the inverse Markovian Mpemba effect). We have found the sufficient condition for a system to have these effects. Such anomalous cooling and heating effects were numerically demonstrated in a minimal three-state model, the Ising model, and Fokker–Planck dynamics. Similar analyses can be used to predict and identify these effects in many other systems.

By generalizing the Mpemba effect, we demonstrate that anomalous cooling and heating effects are expected not only in water, but also in well-studied systems such as the Ising model, and that these counterintuitive effects can be reconciled with nonequilibrium stochastic thermodynamics. Finding the explicit cause of the Mpemba effect in freezing water is still an interesting

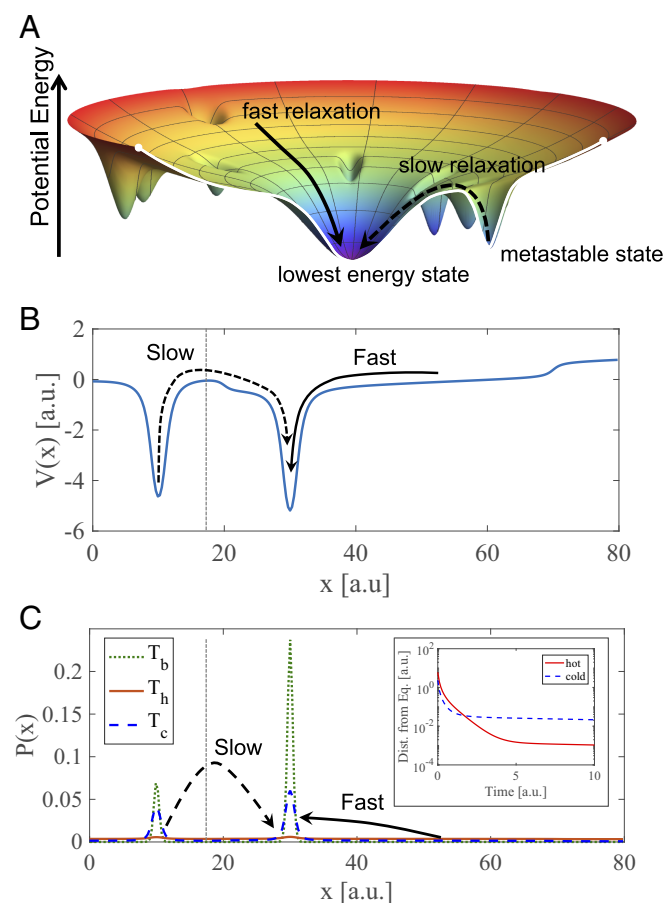


Fig. 5. (A) A schematic sketch of an energy landscape in a high-dimensional configuration space as a 2D energy funnel (plotted using https://oaslab.com/Drawing_funnels.html). The funnel is cut open for illustrative purposes. The solid arrow represents a fast relaxation and the dashed arrow represents a slow relaxation. (B) As an example, we construct a 1D energy landscape and demonstrate the Mpemba effect in the corresponding Fokker–Planck dynamics. The well on the left is a metastable state, whereas the well on the right represents the lowest-energy state. Note that the basin width of the deeper well is significantly larger than that of the shallower well. (C) The Boltzmann distributions at different temperatures. After the quench, both initial distributions relax toward the final equilibrium distribution (dotted green line). Although the initially colder (dashed blue line) system is more populated in the lowest well compared with the initially hot system (solid red line), after a short time of relaxation, the hot system ends up with a higher population in the lowest well due to the fast relaxation from its basin. This grants the initially hot system an advantage over the colder one, and the Mpemba effect occurs.

open problem, which may involve detailed knowledge of the hydrogen bond formation and hydrodynamic currents during cooling. Among the existing explanations of the Mpemba effect in water, convection flow and anomalous relaxation of the hydrogen bonds are manifestly Markovian out-of-equilibrium processes, and hence they might be related to the general mechanism described in this paper.

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